Aravind S

AI Engineer | Generative AI, Agentic AI, LLM Evaluation | Python, AWS

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SUMMARY

AI engineer who builds and ships LLM applications. Built and run thought2build.com, a live multi-provider GenAI product with prompt regression tests in CI, quality gates and tiered model routing. TCS Kochi AI Hackathon winner and national finalist; 5+ years at TCS. Claude Certified Architect – Professional; 2x AWS Certified.

TECHNICAL SKILLS

Generative AI: Claude, OpenAI and Gemini APIs, multi-provider model routing, prompt engineering, RAG (ChromaDB), AI agents, multi-agent orchestration (LangChain), MCP, guardrails, prompt-injection defences
LLM Evaluation & Observability: golden-dataset regression tests, LLM-as-judge, evals in CI, Langfuse, OpenTelemetry, Prometheus, Sentry
Machine Learning: PyTorch, Hugging Face Transformers (BERT, DistilBERT fine-tuning), scikit-learn, pandas, NumPy
Backend & Frontend: Python, FastAPI, Pydantic, SQLAlchemy (async), PostgreSQL, Redis, arq workers, REST APIs, OAuth 2.0 / JWT, TypeScript, React
Cloud & DevOps: AWS (Lambda, API Gateway, S3, ECR, IAM, DynamoDB, SageMaker), Docker, Terraform, GitHub Actions CI/CD, pytest, Playwright

PROFESSIONAL EXPERIENCE

Tata Consultancy Services (TCS) | Kochi, India **Feb 2021 - Present**
Systems Engineer, TCS Digital *(since Feb 2023); previously Assistant Systems Engineer*

AI HACKATHON PROJECTS

SentinelIQ | Winner, TCS Kochi AI Hackathon github.com/Arv-ind-s/sentineliq

- Built a predictive-maintenance app for a simulated 15-machine plant; its scikit-learn models caught 22 of 24 failure events (0.987 balanced accuracy) on a held-out test set.
- Used Gemini only to explain results the models had already computed, with a template fallback, and required a planner's approval before any work order reaches the ERP.

Covenant Radar | Finalist, TCS National Hackathon github.com/Arv-ind-s/CovenantRadar

- Built an early-warning system that flags commercial loans likely to breach their covenants within 30, 60 or 90 days, tested on the real quarterly filings of 24 NSE-listed companies.
- Gated releases on an offline evaluation: 35 hand-written cases across 10 categories, each with a minimum score.

PROJECTS

Thought2Build | Live at thought2build.com | Apr 2026 - Present github.com/rsqrd-labs/thought2build

- Built and shipped solo a full-stack GenAI product that turns a product idea into a spec, plan, test harness and task list for AI coding agents. FastAPI, PostgreSQL, Redis, React/TypeScript; 860+ commits, 260+ test files.
- Set up prompt regression tests in CI (GitHub Actions): a prompt change must bump its version, 28 graders score golden reference projects, and the PR check fails if any score falls below the baseline.
- Added a rule-based quality gate on every generated stage (required sections, requirement coverage, unsafe technology) with one-click regenerate; an LLM judge then adds suggestions in the background.
- Built tiered model routing across Claude, GPT and Gemini: requests start on small models and move up a tier only on failure, and a cheaper route becomes the default only after it passes the golden evaluation.
- Moved GitHub sync and payment processing onto background worker queues (Redis, arq) with automatic retries, so a burst of exports never delays a paid credit grant.

Nexus Research Assistant | Agentic RAG github.com/Arv-ind-s/Nexus-Research-Assistant

- Built a LangChain multi-agent research assistant that answers from a private ChromaDB document store, live web search (Tavily) or both, with a cache that cuts repeat API cost and latency.

Real-Time Content Moderation | Serverless AWS inference github.com/Arv-ind-s/content-moderation-system

- Fine-tuned DistilBERT to flag six kinds of toxic comment (Jigsaw dataset, 159,571 comments) and served it as a serverless AWS API (Lambda container, API Gateway, S3), provisioned with Terraform.

CERTIFICATIONS

Claude Certified Architect – Professional | AWS Certified ML Engineer – Associate | AWS Certified Developer – Associate

EDUCATION

Adi Shankara Institute of Engineering and Technology | *B.Tech, Electrical and Electronics Engineering* 2016 - 2020